



Module 5: Capstone

Practice Exam · Targeted Review · VE Exam
You've done the work — now prove it



The Exam — What to Expect

- 35 multiple-choice questions, one from each group
- 26 correct to pass = 74% — you can miss 9
- Most people finish in 15–20 minutes
- No trick questions — the answer is always there
- Calculators allowed (barely needed)
- VE exam right here at the lab — you already know this room



What to Bring to the VE Exam

- Photo ID (driver's license, passport, state ID)
- Your FRN (FCC Registration Number) if you have one
- A pencil and basic calculator (optional)
- You do NOT bring money — \$35 fee is paid online later
- The fee posts to CORES after VEs submit your application
- You have 10 days to pay it online at fcc.gov/cores



Practice Exam #1

35 questions · 30 minutes · Real conditions



Practice Exam #1 — Instructions

- Work silently and individually — just like the real exam
- Time yourself: 30 minutes maximum
- Circle your answers on the sheet
- When done, put your pencil down
- We'll grade together as a group
- Then we review every question anyone missed
- Focus on WHY the right answer is right



Review Missed Questions

Going through every miss



How to Review

- Go through each missed question
- Don't just say the right letter — explain WHY
- Connect to the module it came from
- If multiple students missed the same question, spend extra time there
- Identify weak subelements per student
- The questions they miss here are the ones they'll miss on the real exam



The Free Points — Know These Cold

Rules & Licensing

- FCC regulates amateur radio
- License: 10 years, 2-year grace (no TX)
- ID: every 10 min + end, in English
- Control operator: ALWAYS required
- 2m calling: 146.520 MHz
- Club station: 4 members minimum
- Emergency: any frequency allowed

Technical & Safety

- $V = I \times R$ (Ohm's Law)
- $P = V \times I$ (Power)
- $300 \div \text{MHz} = \text{wavelength}$
- $468 \div \text{MHz} = \text{dipole length (ft)}$
- +3 dB = double, +10 dB = 10×
- 50 ohms standard impedance
- Power lines = #1 antenna danger
- Rosin-core solder, never acid



Practice Exam #2

20 questions · Focused on weak areas



Practice Exam #2 — Targeted Drill

- 20 questions focused on YOUR weak spots
- Based on what you missed in Practice Exam #1
- Same format — silent, individual, timed
- Grade and review quickly
- This is your final calibration before the real thing
- If you're scoring 80%+ on this, you're passing the exam



You're Ready

Go earn your ticket



What Happens After You Pass

- VEs submit your application to the FCC
- \$35 fee posts to CORES — pay within 10 days at fcc.gov/cores
- Call sign appears in ULS within 1–2 weeks
- You can transmit as soon as it appears in ULS
- Download your license from ULS — print it, frame it



Your First Week as a Licensed Operator

- Pay the \$35 fee immediately when it appears in CORES
- Watch ULS daily for your call sign
- First contact: 146.520 MHz — call CQ!
- Program local repeaters (repeaterbook.com)
- Check in to a net — great practice
- Log your contacts — notebook or digital
- Come back to the lab — Elmer sessions, POTA, and more



Congratulations

- You studied. You drilled. You practiced.
 - You are ready to pass.
 - Not 'if' — WHEN you pass.
 - Welcome to amateur radio.
 - 73 — and we'll see you on the air.
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- TavaOne Education · Pinellas STEM Radio Lab · tavaoneeducation.org